

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import joblib
```

```
data = {
    'YearsExperience': [1.1, 1.3, 1.5, 2.0, 2.2, 2.9, 3.0, 3.2, 3.2, 3.7, 3.9, 4.0, 4.0, 4.1, 4.5],
    'Salary': [39343, 46205, 37731, 43525, 39891, 56642, 60150, 54445, 64445, 57189, 63218, 55794, 56957, 570
}
df = pd.DataFrame(data)
```

```
X = df[['YearsExperience']]
y = df['Salary']
```

```
model = LinearRegression()
model.fit(X, y)
```

▼ LinearRegression ⓘ ?

```
LinearRegression()
```

```
joblib.dump(model, 'linear_model.pkl')
print("Model saved as linear_model.pkl")
```

Model saved as linear_model.pkl

```
y_pred = model.predict(X)
score = r2_score(y, y_pred)
print("Model trained and saved as 'linear_model.pkl'")
print("R2 Score:", round(score, 3))
```

Model trained and saved as 'linear_model.pkl'
R2 Score: 0.702